CLOUD NATIVE HACKATHON

# ConfPilot

## AI Conference Planner

Turn a 300-talk schedule into a personal, clash-free plan in seconds. Just describe yourself.

Built by Sai Ganesh Vuppala

# Big conferences are hard to navigate.

- **Hundreds of talks**
  Many run at the same time in different rooms.
- **Hours of reading**
  Attendees still miss the sessions that matter to them.
- **Double-booking**
  The best talks overlap. Choosing is painful.
- **Wrong level**
  Beginner sessions for experts, deep dives for newcomers.
- **Missed people**
  No easy way to find others working on the same problems.

# Describe yourself. Get a complete plan.

### Personal agenda

Best talks for you, grouped by day, with no time clashes

### AI reasons

Why each talk fits your profile, plus practical tips

### Backup options

One-click swap if a preferred talk is full

### Chat to refine

“Swap day 2 for AI talks” — agenda updates and stays clash-free

### People to meet

Attendees working on similar things, with conversation openers

### Calendar export

Download the full agenda as an .ics file

# From one sentence to a finished agenda

1

## Understand

An LLM reads your profile and extracts a search query and skill level.

2

## Hybrid search

Elasticsearch runs keyword search (BM25) and meaning search (vectors), then fuses the two rankings.

3

## Plan

A planner selects the best non-overlapping talks and keeps a backup for each slot.

4

## Explain

AI writes a short reason for every talk and conversation openers for people you should meet.

5

## Refine

Chat lets you change the agenda. Every edit goes through the same search and clash check.

# Keyword and meaning, together

## KEYWORD SEARCH

BM25 exact matching.
Fast and precise when the talk title already contains the words you care about.

Example: “eBPF security”

## MEANING SEARCH

Dense vector embeddings of talk abstracts.
Finds related talks even when no words are shared.

“save money and help the planet”
→ FinOps, autoscaling, carbon

Results are fused with Reciprocal Rank Fusion. Badges show which search found each talk.

# Built on the tools the tracks require

### Elasticsearch

Stores talks and attendees. Hybrid search is the core of every request.

### OpenTelemetry + Dash0

Every request is traced. Custom metrics track search latency and AI outcomes. Logs are linked to traces.

### Kubernetes + Radar

Full manifest with probes, limits and pod identity, deployed with one script. Radar shows pod health and diagnoses six scripted incidents.

### Gitar + CI

CI runs lint, unit tests and a real Elasticsearch test. Gitar blocked a rule-breaking pull request with 4 findings, then fixed it.

# Designed to keep working

- **AI fallbacks**

  If the primary model is busy or retired, a backup answers. If every AI call fails, rule-based reasons are used.

- **Graceful degradation**

  The page shows a clear notice when AI is unavailable. The agenda is still generated.

- **Health checks**

  Readiness fails when Elasticsearch is unreachable, so Kubernetes stops sending traffic.

- **Validation**

  A single script checks the full stack before use.

# A personal conference plan in seconds

→ Clash-free agenda tailored to your interests and level

→ Clear reasons for every recommended talk

→ Backup options for each time slot

→ Relevant people to meet, with conversation starters

→ Ability to refine the plan in plain English

→ One-click calendar export

# ConfPilot

## Describe yourself.
## Get a smarter conference.

Powered by Elasticsearch hybrid search
Observed with Dash0 · Runs on Kubernetes with Radar · Reviewed by Gitar

`github.com/vuppalasaiganesh/ConfPilot`

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